Censoring in Survival Analysis

Right, left and interval censoring; Type I, II and III designs;
random versus informative censoring

Study notes with biological examples and illustrations

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1 What is censoring?

In survival analysis the outcome of interest is the *time to an event* (death, relapse, infection, recovery, device failure). **Censoring** is present when information on the time to the outcome event is not fully available. A subject is said to be *censored* when the exact event time is unknown, for example because of loss to follow-up, withdrawal, or non-occurrence of the event before the trial ends. We know only that the event time lies in some range.

Censored observations still carry information ("this subject survived at least this long"), so they are kept in the analysis. Discarding them would bias survival estimates downward.

Notation.

- T: true (possibly unobserved) event time; C: censoring time.
- $S(t) = P(T > t)$ survival function; $f(t)$ density; $F(t) = 1 - S(t)$.
- For right-censored data we observe (t_i, δ_i) with $t_i = \min(T_i, C_i)$ and $\delta_i = 1$ if the event was observed, 0 if censored.

2 Classification by what is observed

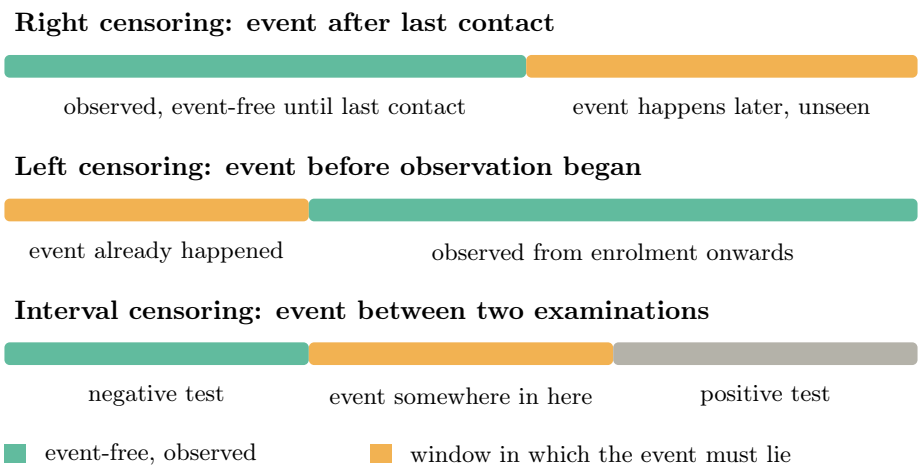


Figure 1: What is known about the event time under right, left and interval censoring.

2.1 Right censoring (the most common form)

Despite continuous monitoring of the outcome, the subject is lost to follow-up or the event does not occur within the study duration. We know only that the event time exceeds the last observation time: $T > C$.

- **Likelihood contribution:** exact event $f(t_i)$; right-censored $S(t_i)$.
- **Biological examples:**
  - *HIV treatment:* patients on antiretroviral therapy followed for viral rebound. Those still suppressed at the end of the study are censored, as is a patient who transfers clinics or dies in an accident when the event of interest is rebound.

- *Tuberculosis:* time to sputum culture conversion, where some patients stop attending clinic before converting.
- *Wildlife ecology:* radio-collared animals followed for time to death, where a collar fails or the animal leaves the study area.

- **Methods:** Kaplan–Meier estimation, log-rank test, Cox regression.

2.2 Left censoring

The event has already occurred before observation started, so $T < C$. We know only that the time is at most C. It is rare in clinical trials but common in epidemiology and laboratory work.

- **Likelihood contribution:** $F(t_i) = 1 - S(t_i)$.
- **Biological examples:**
  - *Age at menarche:* girls already menstruating at the first survey cannot say when it began. We know only that the age is less than their current age.
  - *Malaria in infants:* time to first parasitaemia, where some babies are already parasite-positive at the first clinic visit.
  - *HPV or other infections:* women already HPV-positive at enrolment, so the time of acquisition is unknown.
  - *Assay detection limits:* viral load or toxin concentration below the detection limit; the true value lies between 0 and the limit.

Do not confuse with left truncation. In *left censoring* the subject is in the study and we know the event happened earlier. In *left truncation* (delayed entry) subjects who experienced the event before entry are never observed at all, and must be handled with a different likelihood (risk set starting at entry time).

2.3 Interval censoring

The event occurs within a known interval but the exact time is not observed: $L < T \leq R$.

- **Likelihood contribution:** $S(L_i) - S(R_i)$.
- **Special cases:** right censoring is the case $R = \infty$; left censoring is the case $L = 0$.
- **Biological examples:**
  - *HIV seroconversion:* negative at month 6 and positive at month 12 means infection occurred in $(6, 12]$.
  - *Malaria cohorts:* monthly blood smears detect first infection only at the visit, not on the day it began.
  - *Cancer:* tumour recurrence found at scheduled scans; fracture healing judged on follow-up X-rays.
  - *Dental caries:* onset found at periodic dental examinations.
  - *Breast-cancer radiotherapy:* onset of cosmetic deterioration detected at clinic visits (Finkelstein, 1986).
- **Methods:** Turnbull nonparametric estimator; parametric accelerated failure time models; semiparametric Cox models for interval-censored data.

3 Classification by study design (mechanism of censoring)

All three designs below produce right-censored data in the classical setting; they differ in *what is fixed in advance* and what is random.

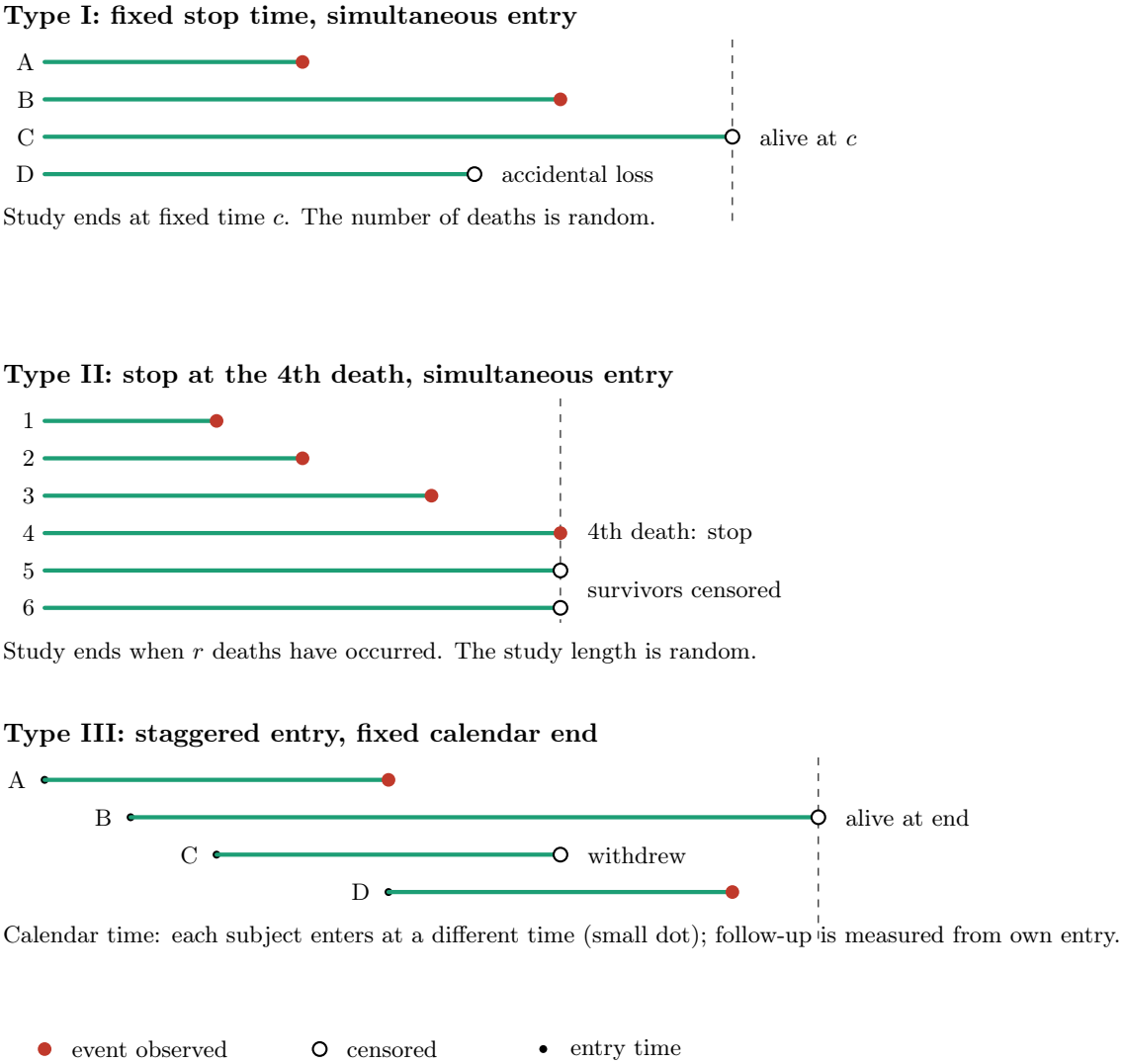


Figure 2: Follow-up lines for the three classical censoring designs.

3.1 Type I censoring (fixed-time censoring)

Design. The time at which the study starts and stops is fixed in advance. All subjects enter at the same time and are followed for the same period c. Those still alive at the end are censored; their survival times are recorded as at least the length of the study period. Animals lost or accidentally killed have survival times from the start of the experiment to the time of loss or death. Subjects that die during the study have exact survival times, from the start to the time of death.

Random quantity: the number of observed events. The censoring time is fixed.

Likelihood:

$$L = \prod_{i=1}^{n} f(t_i)^{\delta_i} \, S(c)^{1-\delta_i}.$$

Biological examples.

- *Toxicology:* 50 mice receive a carcinogen and are observed for exactly 24 weeks. Mice dying earlier have exact times; the rest are censored at week 24.
- *Ecotoxicology:* zebrafish embryos or *Daphnia* exposed to a pesticide for a fixed 96 hours.
- *Vaccine challenge:* animals challenged with a pathogen and observed for a fixed 28 days.
- *Reliability analogue:* a life test of 100 devices run for exactly 1 000 hours.

3.2 Type II censoring (fixed number or proportion of events)

Design. The number (or proportion) r of subjects that must die is fixed in advance. All n subjects start together; the study stops when the r-th death occurs, after which the surviving subjects are sacrificed or censored.

Random quantity: the study duration (time of the r-th failure). The number of events is fixed at r and the number censored is $n - r$.

Likelihood:

$$L \propto \frac{n!}{(n-r)!} \prod_{i=1}^{r} f\big(t_{(i)}\big) \; \big[S\big(t_{(r)}\big)\big]^{\,n-r}.$$

Worked example. Six rats are poisoned and their times of death are observed; the investigator decides to terminate the study when 4 rats are dead. If deaths occur at days 3, 5, 6 and 9, the study ends at day 9 and the remaining 2 rats are censored at day 9. The data are

$$(3,\ 5,\ 6,\ 9,\ 9^{+},\ 9^{+}).$$

Application. Animal experiments and early-phase life testing, where it saves time because the study need not wait for the slowest subject to die. It is rare in human trials, where it is impractical to end everyone's follow-up at a random calendar time.

3.3 Type III censoring (fixed study period, staggered entry)

Design. The calendar start and end of the study are fixed in advance, but subjects enter at different times during the period. For each subject one of three things happens:

(1) the event occurs before the study ends, giving an exact time (entry to event);

(2) the subject withdraws or is lost to follow-up, giving a right-censored time (entry to last contact);

(3) the subject is still event-free at the study end, giving a right-censored time (entry to study end).

Random quantities: both the number of events and the censoring times. The likelihood has the same form as for right censoring, with subject-specific censoring times c_i:

$$L = \prod_{i=1}^{n} f(t_i)^{\delta_i} \, S(c_i)^{1-\delta_i}.$$

Type III is also called *random censoring* or *generalized Type I censoring* in some textbooks.

Worked example. A 24-month trial with enrolment over the first 12 months:

Patient	Entry (month)	Outcome	Time since entry	Status
A	0	Dies at month 10	10	event ($\delta = 1$)
B	3	Alive at study end (month 24)	21^+	censored
C	5	Moves away, last seen month 14	9^+	censored
D	12	Relapses at month 20	8	event ($\delta = 1$)

Survival time is measured from each patient's own entry (study time), not calendar time.

Biological examples.

- *Clinical trial:* breast-cancer patients recruited over 2 years, trial closing on a fixed date. Time to relapse is measured from each patient's enrolment.
- *Cohort studies:* HIV, tuberculosis and cardiovascular cohorts are almost always of this kind.

3.4 Comparison of the three designs

Feature	Type I	Type II	Type III (random)
Entry	Simultaneous	Simultaneous	Staggered
Stopping rule	Fixed time	Fixed number of events	Fixed calendar end date
Censoring time	Same for all (fixed)	Time of r-th failure (random)	Varies by subject (random)
Number of events	Random	Fixed (r)	Random
Typical setting	Animal and laboratory studies	Life testing, animal experiments	Clinical trials, cohort studies

4 Random (non-informative) versus informative censoring

Non-informative (random) censoring assumes the censoring mechanism is unrelated to the underlying event time. Standard methods (Kaplan–Meier, log-rank test, Cox regression) assume this.

Informative censoring means censoring is related to the event time, which requires special methods of analysis.

- *Non-informative example:* a participant relocates for a new job. This is unrelated to prognosis, so standard methods remain valid.
- *Informative example:* patients in a chemotherapy trial withdraw because side effects are making them sicker. Censoring is then related to the event time, and Kaplan–Meier will overestimate survival.
- *Remedies:* sensitivity analyses, competing-risks methods, or inverse-probability-of-censoring weighting.

5 Summary table

Type	What is known	Likelihood term	Biological example
Exact event	$T = t$	$f(t)$	Death of a mouse at week 10
Right censored	$T > C$	$S(C)$	Patient event-free at study end
Left censored	$T < C$	$F(C) = 1 - S(C)$	Girl already menstruating at survey
Interval censored	$L < T \leq R$	$S(L) - S(R)$	HIV seroconversion between visits

6 Practical notes for analysis

- Always record an event indicator δ alongside the observed time.
- Report the reasons for censoring and check the non-informative censoring assumption.
- Handle left truncation (delayed entry) separately from left censoring.
- Right-censored data: Kaplan–Meier, log-rank test, Cox regression. Interval-censored data: Turnbull estimator, parametric models. Parametric models (exponential, Weibull) are fitted by maximum likelihood using the contributions in the summary table.

R sketch.

```
library(survival)
# Right-censored (Type III table): time since entry and status
d <- data.frame(time = c(10, 21, 9, 8), status = c(1, 0, 0, 1))
fit <- survfit(Surv(time, status) ~ 1, data = d)
summary(fit)

# Interval-censored: L = last negative, R = first positive
# (use R = NA for right-censored, L = 0 for left-censored)
# survreg(Surv(L, R, type = "interval2") ~ 1, dist = "weibull")
# library(icenReg); ic_np(cbind(L, R) ~ 0, data = dd)  # Turnbull
```

7 Notes on terminology

- In most textbooks (Klein & Moeschberger; Lawless) Type I means a *fixed* censoring time. Completely random dropout and staggered entry are usually called Type III or *random censoring*. Some course notes group random dropout under Type I; follow your lecturer's definition for examinations.
- "Point censoring" is not a standard textbook name for right censoring.
- These notes were compiled from standard textbook definitions rather than live retrieval, so please check page-level details against the originals below.

References

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